

Secure Programming

CIS-225 30 0

Fall Term 2025-2026 School Year Section 30 0 3.00 Credits 08/25/2025 to 12/12/2025

Modified 10/24/2025

Course Description

This course will introduce a number of security fundamentals, including physical security risks, social engineering and cyber vulnerabilities. The course will explore software exploitation and the approaches that malicious users will take to exploit programs, as well as mitigation techniques to prevent incidents.

Prerequisite: CIS 131, CIS 196

Outcomes

Critical Thinking

- Document potential vulnerabilities and weaknesses of software and network architecture to highlight key areas for defense and monitoring.
- Research and present emerging exploits and assess the CVSS scores of real world and theoretical vulnerabilities to present to stakeholders.
- Analyze and fix software vulnerabilities in code and implement best practices for data assurance within a secure and user friendly UX.

Communication

- Student communicates high level concepts of technology in a manner accessible by a non-technical audience.
- Student outlines potential threats and mitigations including technical details in a clear and concise manner for a technical audience.

Professionalism

- Utilizes tools to create clean, professional visuals/presentation materials.

Technical Skills

- Implement HTTPS/TLS in a secure web app.

Course Materials

N/A

All materials for this course are available within the course page.

Deliverables

Homework

Weekly assignments will be used to track attendance. They will be a completion grade, and are intended as student self assessments. The instructor will not make corrections to the homework. It is intended that students study the material provided until they have a good understanding of it.

Unit Project

There are 3 units in this course, each with a Unit Project. Unit Projects will be the assessments for this course, where you will create an application to demonstrate skills, and then present the application to prove your skills. Presentations may be recorded if not attending class on presentation day.

In addition to the presentation, your instructor may require an interview on the Unit Projects to answer follow-up questions and gauge your understanding of the material.

Within Unit Projects, students will be assessed on competencies, which will be used to determine their Project Competencies - Approaching, Proficient, and Mastered scores.

Evaluation Procedures and Grading

Criteria

Please Note: The intent of this grading system is for students to understand each topic in the course in order to pass. Completion of the homework and demonstrated mastery of the course topics is encouraged to raise your grade beyond a C, presuming that you are meeting proficiency in all of the competencies.

Type	Weight	Topic	Notes
Project Competencies - Approaching	40%		Student achieves at least a 2/4 on every project competency
Project Competencies - Proficient	30%		Student achieves at least a 3/4 on every project competency

Type	Weight	Topic	Notes
Project Competencies - Mastery	10%		Each 4/4 score on project competencies, students will receive partial credit towards this score. Example: a student achieving 4/4 on half of the competencies will receive 5% of their final grade weight from this.
Participation/Homework	20%		Homework assignments weighted equally will give partial credit towards this score.

Breakdown

This breakdown is a model/example, not a hard rule for the final grading. Final grading will depend on the grade weighting as defined in the criteria.

Grade	Range	Notes
A	90%-100%	For an A, it is expected that students attend every class session , complete every competency with a score of 3/4 or higher, complete all homework, and completing some competencies with a score of 4/4.
B	80%-89%	For a B, it is expected that students achieve a 3/4 on every competency and also complete most homework assignments, and possibly receive a few 4/4 scores on competencies.
C	70%-79%	C is the minimum passing grade for this course. To receive a C in this course, a student must complete every competency on each assessment with a 3/4 or higher.
D	60% - 69%	Students participating completing all homework but not passing every competency with a 3/4 or higher may expect to receive a D in this course, if they complete every competency with a 2/4.



Course Outline

When	Topic	Notes
Unit 1 Weeks 1-3	Risk Assessment	• Data Flow Diagrams • Monitoring and Logging • Reporting Vulnerabilities • Determining Potential Threats
Unit 2 Weeks 5-8	Vulnerability Classification and Analysis	• CVEs and Exploit Reporting • CWEs and Vulnerability Classification • CVSS Scoring • OWASP Top 10

When	Topic	Notes
Unit 3 Weeks 8-16	Vulnerability Mitigation	• Privileges • Social Engineering • Cryptography • Injection • Session Management • Supply Chain Security